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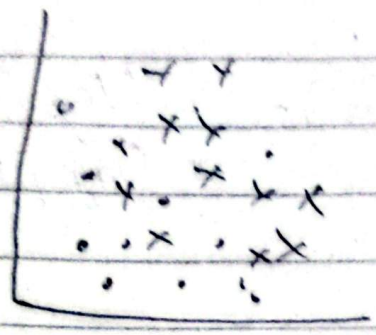
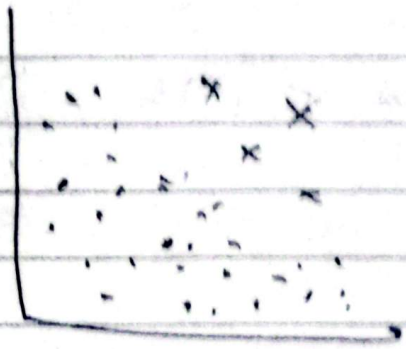
- How much difficult to predict?
  Difficulty ↑ → Entropy ↑



High entropy.



Low entropy.

Entropy:

$$E(S) = \sum_{i=1}^{c} (-P_i \log_2 P_i)$$

where, P_i = Frequentist probability of an element/ class 'i' in our data

For eg: 2 class labels yes or no.

$$E(S) = -P_{yes} \log_2 (P_{yes}) - P_{no} \log_2 (P_{no}).$$

eg: $E([7+, 7-]) = -\frac{7}{14} \cdot \log_2\left(\frac{7}{14}\right) - \frac{7}{14} \log_2\left(\frac{7}{14}\right)$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

iq	gpa	placement
78	7.8	Yes
80	8	Yes
60	6.8	NO

$\}2$, $\}1$, $\}3$

$$E(\mathcal{D}) = -P_{yes}\log_2(P_{yes}) - P_{no}(\log_2(P_{no}))$$
$$= -\frac{2}{3}\log_2\left(\frac{2}{3}\right) - 1\left(\log_2\left(\frac{1}{3}\right)\right)$$

If all labels are same either all yes or all no, then the entropy will be zero

<u>Observation:</u>

- More the uncertainity more is entropy.
- For a 2 class problem the min entropy is 0 and the max is 1.
  if no. of yes = no. of no.
- For more than 2 classes the min entropy is 0 but the max can be greater than 1.
- Use either $\log_2$ or $\log_e$

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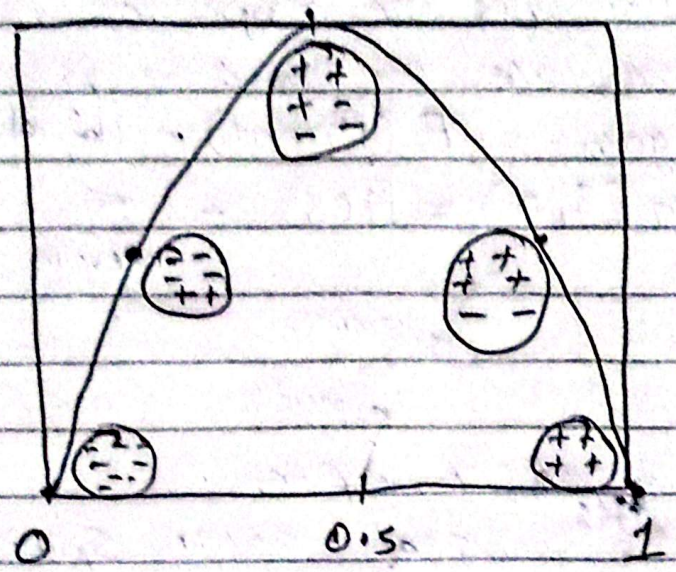
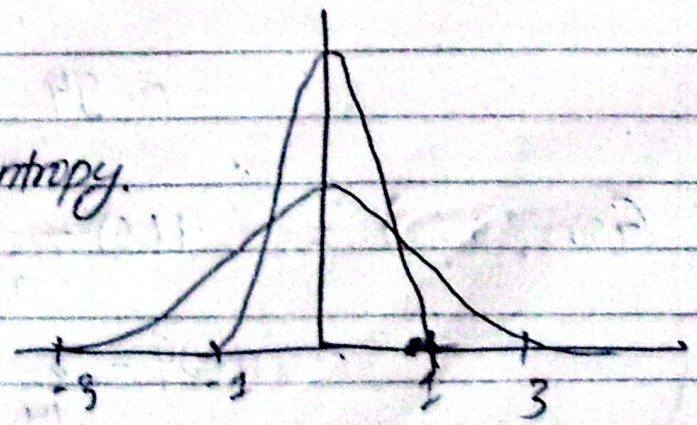
Entropy Vs Probability.



Entropy for continuous variable.

① Prices 3.5 5.6 7.3 - - .
② Prices 4.6 6.5 12.8 - - .

- ~~Do~~ which of the above datasets have higher entropy?
→ Do KDE plot.

low peak → high entropy.